THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 64	Sum of first n odd numbers $= n^2 = 64$
2	**(a)** 4	$54 = 2 \times 3^3$; 2×2 more
3	**(d)** 6	$64 = 2^6$
4	**(d)** 444	Add the values, subtracting a smaller symbol that comes before a larger one
5	**(a)** kite	Kite
6	**(b)** 13 cm	$\sqrt{l^2 + b^2}$
7	**(a)** 12	Greatest common factor
8	**(c)** 2000	$(a+b)(a-b)$
9	**(b)** 20	Product of extremes = product of means: $3x = 60$
10	**(b)** 2 : 5	Divide both by HCF 18
11	**(b)** interest is also earned on earlier interest	Compounding
12	**(b)** ₹ 800	$SI = \frac{PRT}{100}$
13	**(c)** Yes	$6^2 + 8^2$ = 10^2
14	**(b)** 10 cm	$\sqrt{6^2 + 8^2} = 10$
15	**(b)** 4 cm	200 cm / 50
16	**(b)** square	Same as a face
17	**(a)** increases by 5	Shift
18	**(d)** odd	$2n + 1$
19	**(d)** 24 cm²	Other diagonal 8
20	**(d)** perimeter	Perimeter is not a measure of area

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	3×10^8 m/s (2).	2
22	$7 \times 64,\ 3 \times 8,\ 4 \times 1$ (2).	2
23	$1001 = 7 \times 11 \times 13$, so yes to all (2).	2
24	When one quantity increases, the other increases in the same ratio (1); cost and quantity (1).	1+1
25	The diagonal of a rectangle produces an area equal to the sum of the areas produced by its length and breadth (2).	2
26	₹ 240 (2).	2
27	On a number line with dots as weights, the mean is where the line balances (2).	2
28	15 m (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	LCM $= 24 = 2^3 \times 3$ (1); need $2^4 \times 3^2$ (1); least square $= 144$ (1).	1+1+1
30	Both diagonals are hypotenuses of right triangles with legs l and b (2); so both $= \sqrt{l^2 + b^2}$ (1).	2+1
31	$6x^2 - 15x + 4x - 10 = 6x^2 - 11x - 10$ (3).	3
32	$R = \frac{100 \times 1920}{8000 \times 3}$ (2); 8% (1).	2+1
33	Square, circle, circle with centre, square with diagonals (3).	3
34	$a + 2 \times 6 + c = 25$ (1.5); $c = 6$ (1.5).	1½+1½
35	$\sqrt{225 - 81} = 12$ m (3).	3

	OR: Side 17 cm (2); perimeter 68 cm (1).	2+1
36	$196 - 36 = 160$ cm² (3).	3
	OR: Area $= \frac{1}{2} \times 25 \times 14 = 175$ m² (2); cost ₹ 2100 (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 1.4×10^9 and 3.29×10^6 (2). (ii) ≈ 426 per km² (2). (iii) City is about 26 times denser (1).	2+2+1
	OR: (i) 0.00047 (1). (ii) 9.2×10^{-7} (2). (iii) $35000 = 3.5 \times 10^4$ (2).	1+2+2
38	(i) $25000 \times 1.1664 = 29160$ (2). (ii) ₹ 4160 (1). (iii) SI ₹ 4000; difference ₹ 160 (2).	2+1+2
	OR: (i) MP 1.4, SP 1.05, profit 5% (3). (ii) ₹ 525 (2).	3+2
39	(i) Mark each value on a number line with a dot (2). (ii) Mean 8, median 8 (3).	2+3
	OR: (i) $208/7 \approx 29.7$°C (2). (ii) Wednesday (1). (iii) Range 6; rises then falls (2).	2+1+2
40	(i) Equal bases and common height (3). (ii) 40 cm² (2).	3+2
	OR: (i) $10 \times 6 = 60$ m² (2). (ii) $96 - 60 = 36$ m² (1). (iii) ₹ 1980 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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